

Subject Description Form

Subject Code	DSAI4101
Subject Title	Reinforcement Learning
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	<u>Pre-requisite</u> 1. Probability and Statistics Statistical Inference (AMA3640) AND 2. Linear Algebra (one of the following subjects) Linear algebra (AMA1751); Calculus and Linear Algebra (AMA1007); Calculus and Linear Algebra (AMA10071); Basic Mathematics II - Calculus and Linear algebra (AMA1120) AND 3. Python Programming (one of the following subjects) Principles of Programming (AMA2222); Computational Thinking and Problem Solving (COMP1010)
Objectives	1) Introduce the fundamental principles of Reinforcement Learning (RL), including the agent-environment interaction and the formulation of problems as Markov Decision Processes (MDPs). 2) Present the main families of RL algorithms, including value-based and policy-based approaches. 3) Introduce advanced concepts such as function approximation with deep neural networks (Deep RL), the exploration-exploitation trade-off, and policy gradient methods. 4) Provide hands-on experience using modern RL libraries (e.g., Gymnasium, PyTorch) to implement, train, and evaluate agents for solving sequential decision-making problems in domains like game playing.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) formulate a real-world sequential decision-making problem as a Markov Decision Process (MDP) and understand the core concepts of states, actions, rewards, and policies; (b) implement and differentiate between key RL algorithms, including Q-learning, SARSA, and basic policy gradient methods, and understand their theoretical underpinnings like the Bellman equations; (c) understand and apply deep learning techniques for function approximation in large state spaces (Deep Reinforcement Learning), including Deep Q-Networks (DQN);

	(d) master software tools for building and training RL agents (e.g., Gymnasium, PyTorch) and apply these models to solve complex problems, analyze agent performance, and interpret the results.
Subject Synopsis/ Indicative Syllabus	Foundations of Reinforcement Learning (4L+2T) <i>Introduction to RL: agent, environment, state, action, reward; Markov Decision Processes (MDPs): transition probabilities, policy, value function; The Bellman equations: Bellman expectation and optimality equations.</i> Dynamic Programming and Model-Based RL (4L+2T) <i>Planning with a known model; Policy Iteration; Value Iteration; Introduction to Model-Based RL: learning a model of the environment.</i> Model-Free Prediction and Control (6L+3T) <i>Monte Carlo (MC) methods for prediction and control; Temporal-Difference (TD) learning: TD(0), SARSA, Q-learning; The exploration vs. exploitation trade-off: epsilon-greedy, UCB.</i> Function Approximation and Deep RL (6L+3T) <i>Value function approximation with linear models and neural networks; Deep Q-Networks (DQN) and its variants (e.g., Double DQN); Experience replay.</i> Policy Gradient Methods (4L+2T) <i>The Policy Gradient Theorem; REINFORCE algorithm; Introduction to Actor-Critic methods.</i> Case studies (2L+1T) <i>RL libraries: Gymnasium, Stable-Baselines3, PyTorch/TensorFlow; Applications in classic control and game playing (e.g., Atari); Case studies on training and deploying RL agents.</i>
Teaching/Learning Methodology	The subject will be delivered through a combination of lectures and tutorials. Lectures will cover the theoretical foundations of Reinforcement Learning, from MDPs to advanced deep RL algorithms. Tutorials will focus on problem-solving, mathematical derivations, and guiding students to use Python-based libraries like Gymnasium and PyTorch to implement, train, and analyze RL agents in various simulated environments.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed			
			a	b	c	d
	1. Assignments	20%	✓	✓		✓
	2. Project	30%	✓	✓	✓	✓
	3. Examination	50%	✓	✓	✓	
	Total	100%				
Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:						
This subject balances theoretical understanding with practical implementation skills. Assignments (20%) form a core part of the continuous assessment, ensuring students can apply theoretical concepts (ILO 1, 2) to practical coding challenges and problem-solving using modern RL libraries (ILO 4). The Project (30%) is a capstone task where students will select a challenging RL problem, formulate it as an MDP, choose and implement an appropriate algorithm (often a Deep RL method), and present their findings. This assesses all learning outcomes (ILO 1-4). The Final Examination (50%) is a written exam designed to assess the students' grasp of the fundamental theories, algorithms, and mathematical principles underlying Reinforcement Learning (ILO 1, 2, 3), ensuring a solid theoretical foundation.						
Student Study Effort Required	Class contact:					
	▪ Lectures/Tutorials				39 Hrs.	
	Other student study effort:					
	▪ Assignments/Projects				36 Hrs.	
	▪ Self-study				30 Hrs.	
	Total student study effort				105 Hrs.	
Reading List and References	Textbooks:					
	Sutton, Richard S., and Andrew G. Barto.	Reinforcement Learning: An Introduction	MIT press, 2018 (2nd edition)			